BGP Configuration

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Purpose:

The purpose of this lab was to familiarize ourselves with BGP and learn the basic configuration for ebgp. We chose to also use ISIS, so we learned how to configure ISIS for ipv4 and ipv6.

Background:

BGP was designed to exchange routes throughout different autonomous systems (AS) on the internet. We specifically used External BGP or EBGP which exchanges these routes outside of autonomous systems. BGP can also run between two different peers in the same autonomous system. This is called internal BGP or IBGP.
The internet is made up of thousands of smaller networks called autonomous systems. A small company may have pool or routers that may form one autonomous system, and a larger company may have a large pool of routers that forms more than one autonomous system. Hundreds of thousands of these autonomous systems form the internet. EBGP routes between these ASs and chooses the best route to take to route to the destination. ASs are usually managed by ISPs and are registered through IANA. While EBGP routes between ASs, IBGP routes inside the AS. IBGP is an internal version of BGP which can be thought of as more of a local version. IBGP is not required inside of networks when using EBGP.

BGP tries to find the most optimal path to route traffic through. BGP allows for a level of customization when configuring to get the most efficient configuration for the customer's needs. Hop count is not the only metric that BGP considers when choosing a route; weight and local preference are common and easy-to-configure attributes to modify BGP. Our group chose to use these because they are easier to configure on a simplified network with only one possible route. Larger networks become much more complicated to pick the best route, and sometimes another route that is still good is picked.

While BGP is very useful to route across the entire internet, it also poses some threats and vulnerabilities. There have been several people who have accidentally advertised incorrect BGP routes and have shut down the entire Internet for several hours. In 2004 a Turkish ISP accidentally routed the entire internet to themselves and shut the internet down for a few hours. A Pakistani ISP similarly wanted to block YouTube for its users but accidently spread the route to the entire internet blocking YouTube for a few hours. An ISP in Pennsylvania routed traffic in versions ASs to themselves causing some of the internet to go down for several hours. There have also been attacks on the internet through BGP called a BGP hijacking. Hackers can intentionally route information from the entire internet to themselves. In 2019, a group of hackers stole traffic intended for amazon and intercepted the traffic routing it to themselves. They ultimately stole around $100,000. While there has been security patches that have been released to hopefully prevent these attacks, not all ISPs use it and BGP is still vulnerable to these attacks although it is more difficult.

ISIS is a routing protocol that is like routing protocols like OSPF. It dynamically creates routes when a new device is added to the network and rather than having to create a new route for every device when one device is added, you can just configure a routing protocol like ISIS on the device that you added to your network. This saves massive amounts of time and allows for massive network expansion that would not be

possible with just static routes. The routing protocol will automatically redistribute the protocol across the network. ISIS is quite useful and is widely used for enterprise networks even more than OSPF. ISIS functions at layer 2 rather than layer 3 which basically means that it does not need an IP address to function. Because it does not need an IP address, it is both more secure and cannot go down when the IP address is broken. ISIS is basically immune to IP address attacks. ISIS supports both ipv4 and ipv6 with the same version. Unlike OSPF where OSPFv2 is used for ipv4 and OSPFv3 is required for ipv6, ISIS just uses one version. Overall, ISIS is more suited for larger enterprise networks in large companies.
Eigrp is another routing protocol that we used during the BGP lab. We decided to put EIGRP in the middle of our topology over ISIS because we are more familiar with EIGRP and have configured it many times. EIGRP, while not as simple as OSPF, is still fairly simple compared to other routing protocols like ISIS. EIGRP is cisco's proprietary, and while it is somewhat open sourced now, it was developed by Cisco. EIGRP is able to do something called unequal cost load balancing. This is when you have two different routes at different costs. EIGRP is able to load balance between them, sending different amounts of traffic through the different routes that it load balances. While there are some benefits to EIGRP, it is not the most used routing protocol in enterprise networks, but it is still good to know.

Lab Summary

We used external BGP to connect OSPF, EIGRP, and ISIS together. Our goal was to be able to ping from one side of the topology to the other, which would show that we both knew how to set up all the routing protocols and that we knew how to set up external BGP. We were also required to set up both ipv4 and ipv6 as ipv6 configuration for BGP is different than for ipv4. We also decided to add local preference and weight which can help the switch determine which route to take although our topology only had one route that a packet could take through the network.

Lab Commands

Redistribute: redistributes the BGP route across the network.

Address-family: separates ipv4 and ipv6 configurations into two sections

BGP Neighbor: configures the BGP neighbor. This is the BGP connection that links the two autonomous systems together.

BGP Network: Configures the BGP network when talking about BGP.

Router BGP: enables the BGP routing process

Show IP BGP neighbors: show information about BGP connections to neighbors

Net: sets a net address for ISIS. This is like a router id and is needed in order for ISIS to work.

Switch 1 (MLS switch)

```
version 17.6
 service timestamps debug datetime msec
 service timestamps log datetime msec
 ! Call-home is enabled by Smart-Licensing.
 service call-home
 platform punt-keepalive disable-kernel-core
 hostname Switch
 vrf definition Mgmt-vrf
 address-family ipv4
 exit-address-family
 address-family ipv6
 exit-address-family
 no aaa new-model
 switch 1 provision c9200l-24t-4g
 ip routing
 login on-success log
 ipv6 unicast-routing
 crypto pki trustpoint SLA-TrustPoint
 enrollment pkcs12
 revocation-check crl
 crypto pki trustpoint TP-self-signed-1225728909
 enrollment selfsigned
 subject-name cn=IOS-Self-Signed-Certificate-1225728909
 revocation-check none
 rsakeypair TP-self-signed-1225728909

 crypto pki certificate chain SLA-TrustPoint
 certificate ca 01
   30820321 30820209 A0030201 02020101 300D0609 2A864886
F70D0101 0B050030
   32310E30 0C060355 040A1305 43697363 6F312030 1E060355
04031317 43697363
   6F204C69 63656E73 696E6720 526F6F74 20434130 1E170D31
33303533 30313934
   3834375A 170D3338 30353330 31393438 34375A30 32310E30
0C060355 040A1305
   43697363 6F312030 1E060355 04031317 43697363 6F204C69
63656E73 696E6720
   526F6F74 20434130 82012230 0D06092A 864886F7 0D010101
05000382 010F0030
```

```
   82010A02 82010100 A6BCBD96 131E05F7 145EA72C 2CD686E6 
17222EA1 F1EFF64D
   CBB4C798 212AA147 C655D8D7 9471380D 8711441E 1AAF071A 
9CAE6388 8A38E520
   1C394D78 462EF239 C659F715 B98C0A59 5BBB5CBD 0CFEBEA3 
700A8BF7 D8F256EE
   4AA4E80D DB6FD1C9 60B1FD18 FFC69C96 6FA68957 A2617DE7 
104FDC5F EA2956AC
   7390A3EB 2B5436AD C847A2C5 DAB553EB 69A9A535 58E9F3E3 
C0BD23CF 58BD7188
   68E69491 20F320E7 948E71D7 AE3BCC84 F10684C7 4BC8E00F 
539BA42B 42C68BB7
   C7479096 B4CB2D62 EA2F505D C7B062A4 6811D95B E8250FC4 
5D5D5FB8 8F27D191
   C55F0D76 61F9A4CD 3D992327 A8BB03BD 4E6D7069 7CBADF8B 
DF5F4368 95135E44
   DFC7C6CF 04DD7FD1 02030100 01A34230 40300E06 03551D0F 
0101FF04 04030201
   06300F06 03551D13 0101FF04 05300301 01FF301D 0603551D 
0E041604 1449DC85
   4B3D31E5 1B3E6A17 606AF333 3D3B4C73 E8300D06 092A8648 
86F70D01 010B0500
   03820101 00507F24 D3932A66 86025D9F E838AE5C 6D4DF6B0 
49631C78 240DA905
   604EDCDE FF4FED2B 77FC460E CD636FDB DD44681E 3A5673AB 
9093D3B1 6C9E3D8B
   D98987BF E40CBD9E 1AECA0C2 2189BB5C 8FA85686 CD98B646 
5575B146 8DFC66A8
   467A3DF4 4D565700 6ADF0F0D CF835015 3C04FF7C 21E878AC 
11BA9CD2 55A9232C
   7CA7B7E6 C1AF74F6 152E99B7 B1FCF9BB E973DE7F 5BDDEB86 
C71E3B49 1765308B
   5FB0DA06 B92AFE7F 494E8A9E 07B85737 F3A58BE1 1A48A229 
C37C1E69 39F08678
   80DDCD16 D6BACECA EEBC7CF9 8428787B 35202CDC 60E4616A 
B623CDBD 230E3AFB
   418616A9 4093E049 4D10AB75 27E86F73 932E35B5 8862FDAE 
0275156F 719BB2F0
   D697DF7F 28
         quit
 crypto pki certificate chain TP-self-signed-1225728909
 certificate self-signed 01
   30820330 30820218 A0030201 02020101 300D0609 2A864886 
F70D0101 05050030
   31312F30 2D060355 04031326 494F532D 53656C66 2D536967 
6E65642D 43657274
   69666963 6174652D 31323235 37323839 3039301E 170D3235
```

31313132 31333435
   31385A17 0D333531 31313231 33343531 385A3031 312F302D
06035504 03132649
   4F532D53 656C662D 5369676E 65642D43 65727469 66696361
74652D31 32323537
   32383930 39308201 22300D06 092A8648 86F70D01 01010500
0382010F 00308201
   0A028201 0100A32A AE556A02 4609C292 12AB62F1 B38BA9F8
34126673 59A62B1C
   6D9B9749 91A05D31 1E281EB3 A93BAA9E 6F5876FC BA24E4BC
6165A23F F0B67687
   DC173028 A945F964 43ABEACA 6EC84B70 F48D92D7 B438FB25
1A51FF3D 42BF4B23
   B3035E9B 60E1874A EF9A4971 23E8B4EB 0477AFBD 085CF381
1AEE23BC B31DF4EC
   16196BF3 55850E93 86D83A69 C995F7BE 9784886E 985A167D
16A2127B 15977F25
   8DA85632 10D6EDA8 455CE081 3446C8E6 3A700EEC 219DC4F8
B1A26044 21D4ABFA
   252CCDB7 E5469CAB 9352743E 17CB8D6B C8319E12 9E207945
BCA580AA 2E4E2CA3
   C0959345 B1F7A5A8 688EB8D8 CC10B389 4BA33D2C 28842522
F03A9550 A76D5F1E
   7FB46D8C DC1B0203 010001A3 53305130 0F060355 1D130101
FF040530 030101FF
   301F0603 551D2304 18301680 14ADFC30 6356081D 6242602C
C242E6C6 7CB4737F
   FF301D06 03551D0E 04160414 ADFC3063 56081D62 42602CC2
42E6C67C B4737FFF
   300D0609 2A864886 F70D0101 05050003 82010100 3C882B2C
991EB342 E92F6BA8
   7528F5CA 5ED6BF61 00CF8E09 CA36FA68 E66B13A7 66773A3C
AA783281 B7D3BDD3
   13B00175 7FE9AC1C 9CC012D0 42D2A65E F2885F8A 7D240E62
65C01B0C F1B25FB7
   F96E6669 AFBCA419 75BBF286 155C6FF4 DB8B6E62 952AB616
D2A99819 993921F6
   C9588F5B 04FA92A2 7CD6A0CB 557A6E37 18B60260 D0C69E56
ABBDD1DF 1738C73E
   98A8BF94 1F66BB32 B9C0B3C7 5B2182A8 0E6CF005 C3818B22
CEF4D79D 965CBE68
   7ADC385B AB1C14DA FD3FA777 FC5D2B8A 9D9B8FAB ADCA4D76
6FA18562 41A11827
   7AAE51D0 9E52CD4C DC8E3F42 9D8957A5 6DD7FFA6 42B1496D
FE38B44A 05B7AEAF
   14E32AD7 37E07AF8 43117749 809D1792 77E145BF
quit

```
 license boot level network-essentials addon dna-essentials

 diagnostic bootup level minimal

 spanning-tree mode rapid-pvst
 spanning-tree extend system-id
 memory free low-watermark processor 10633

 redundancy
mode sso
 transceiver type all
 monitoring

 class-map match-any system-cpp-police-ewlc-control
   description EWLC Control
 class-map match-any system-cpp-police-topology-control
   description Topology control
 class-map match-any system-cpp-police-sw-forward
   description Sw forwarding, L2 LVX data packets, LOGGING,
Transit Traffic
 class-map match-any system-cpp-default
   description EWLC data, Inter FED Traffic
 class-map match-any system-cpp-police-sys-data
   description Openflow, Exception, EGR Exception, NFL Sampled
Data, RPF Failed
 class-map match-any system-cpp-police-punt-webauth
   description Punt Webauth
 class-map match-any system-cpp-police-l2lvx-control
   description L2 LVX control packets
 class-map match-any system-cpp-police-forus
   description Forus Address resolution and Forus traffic
 class-map match-any system-cpp-police-multicast-end-station
   description MCAST END STATION
 class-map match-any system-cpp-police-high-rate-app
   description High Rate Applications
 class-map match-any system-cpp-police-multicast
   description MCAST Data
 class-map match-any system-cpp-police-l2-control
   description L2 control
 class-map match-any system-cpp-police-dot1x-auth
   description DOT1X Auth
 class-map match-any system-cpp-police-data
   description ICMP redirect, ICMP_GEN and BROADCAST
 class-map match-any system-cpp-police-stackwise-virt-control
   description Stackwise Virtual OOB
 class-map match-any non-client-nrt-class
 class-map match-any system-cpp-police-routing-control
```

```
  description Protocol snooping
 class-map match-any system-cpp-police-protocol-snooping
  description Protocol snooping
 class-map match-any system-cpp-police-dhcp-snooping
  description DHCP snooping
 class-map match-any system-cpp-police-ios-routing
  description L2 control, Topology control, Routing control, 
Low Latency
 class-map match-any system-cpp-police-system-critical
  description System Critical and Gold Pkt
 class-map match-any system-cpp-police-ios-feature
  description 
ICMPGEN,BROADCAST,ICMP,L2LVXCntrl,ProtoSnoop,PuntWebauth,MCASTDa
ta,Transit,DOT1XAuth,Swfwd,LOGGING,L2LVXData,ForusTraffic,ForusA
RP,McastEndStn,Openflow,Exception,EGRExcption,NflSampled,RpfFail
ed

 policy-map system-cpp-policy

 interface Loopback0
 ip address 192.168.10.1 255.255.255.252
 ip ospf 1 area 1
 ipv6 address 2001:DB8:ACAD:10::1/120
 ipv6 ospf 1 area 1

 interface GigabitEthernet0/0
 vrf forwarding Mgmt-vrf
 no ip address
 shutdown

 interface GigabitEthernet1/0/1
 no switchport
 ip address 192.168.1.1 255.255.255.0
 ip ospf 1 area 1
 ipv6 address 2001:DB8:ACAD:1::1/64
 ipv6 ospf 1 area 1

 interface GigabitEthernet1/0/2
 interface GigabitEthernet1/0/3

 interface GigabitEthernet1/0/4
 interface GigabitEthernet1/0/5

 interface GigabitEthernet1/0/6

 interface GigabitEthernet1/0/7
```

```
interface GigabitEthernet1/0/8
interface GigabitEthernet1/0/9

interface GigabitEthernet1/0/10

interface GigabitEthernet1/0/11

interface GigabitEthernet1/0/12
interface GigabitEthernet1/0/13

interface GigabitEthernet1/0/14

interface GigabitEthernet1/0/15

interface GigabitEthernet1/0/16

interface GigabitEthernet1/0/17

interface GigabitEthernet1/0/18

interface GigabitEthernet1/0/19

interface GigabitEthernet1/0/20

interface GigabitEthernet1/0/21

interface GigabitEthernet1/0/22

interface GigabitEthernet1/0/23

interface GigabitEthernet1/0/24

interface GigabitEthernet1/1/1

interface GigabitEthernet1/1/2
interface GigabitEthernet1/1/3

interface GigabitEthernet1/1/4
interface Vlan1
no ip address

router ospf 1
ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
```

```
 ipv6 router ospf 1

 control-plane
 service-policy input system-cpp-policy


 line con 0
 stopbits 1
 line aux 0
 line vty 0 4
 login
 transport input ssh
 line vty 5 15
 login
 transport input ssh

 call-home
 If contact email address in call-home is configured as sch-
smart-licensing@cisco.com
 the email address configured in Cisco Smart License Portal will
be used as contact email    address to send SCH notifications.
 contact-email-addr sch-smart-licensing@cisco.com
 profile "CiscoTAC-1"
   active
   destination transport-method http
```

Router 1

```
version 16.9
 service timestamps debug datetime msec
 service timestamps log datetime msec
 platform qfp utilization monitor load 80
 platform punt-keepalive disable-kernel-core
 hostname router1
 boot-start-marker
 boot-end-marker
 vrf definition Mgmt-intf
 address-family ipv4
 exit-address-family
 address-family ipv6
 exit-address-family
 no aaa new-model
 login on-success log
 subscriber templating
 vtp domain cisco
 vtp mode transparent
 ipv6 unicast-routing
 multilink bundle-name authenticated
```

```
 crypto pki trustpoint TP-self-signed-48848711
 enrollment selfsigned
 subject-name cn=IOS-Self-Signed-Certificate-48848711
 revocation-check none
 rsakeypair TP-self-signed-48848711

 crypto pki certificate chain TP-self-signed-48848711
 certificate self-signed 01
   3082032C 30820214 A0030201 02020101 300D0609 2A864886
F70D0101 05050030
   2F312D30 2B060355 04031324 494F532D 53656C66 2D536967
6E65642D 43657274
   69666963 6174652D 34383834 38373131 301E170D 32353130
32373134 33383430
   5A170D33 30303130 31303030 3030305A 302F312D 302B0603
55040313 24494F53
   2D53656C 662D5369 676E6564 2D436572 74696669 63617465
2D343838 34383731
   31308201 22300D06 092A8648 86F70D01 01010500 0382010F
00308201 0A028201
   0100AE2C F0365BA7 373E645D 1D0B8492 0A92BCE0 3E3E681D
C2C7ECD5 6F61C838
   4479427A A0F7FCFF 7CCDE7E0 26ADC798 D5C0807E A6B7F5BC
3C1AF7E4 E510EDA2
   FBBA972C F6D59704 AC074A7A 38B941FC 48F3F0B6 666061B0
1B48834C E8249238
   2784F27A F2F07DE4 BF92B2C3 05F99D65 C987908D A9CF153E
A40FD4A2 25AA80F2
   D96DE50D 8DC98B3E D554835F 9A3A00A5 0E22EC88 E96793BF
8647917B 507D45D2
   F915C2E8 DB198CED D5B4AB96 1FD9765D 41B505FB 28B3FCE4
C80BEF86 AB707EE5
   CB574CEF 98782D5B 0B6F6B33 89699DFA A9010EC6 320129FB
B38D7137 90DB4FED
   11E67BE6 1EF50063 4A717219 3158E651 E1E037CE B29A35B0
07F312E9 F8351071
   4DD70203 010001A3 53305130 0F060355 1D130101 FF040530
030101FF 301F0603
   551D2304 18301680 14D6042B FA5172BA 76C41C78 526D3A54
99828397 90301D06
   03551D0E 04160414 D6042BFA 5172BA76 C41C7852 6D3A5499
82839790 300D0609
   2A864886 F70D0101 05050003 82010100 25EB65C1 63316592
42061D57 679578D1
   7A22C229 B4CC8D5A AAA9E0F7 1ED91562 63231E21 5105C2C7
29DE698A 5CA396B0
   59BC4661 28BF544A 978FA5D4 B98ACE5D 1C32AE2B 9781C9DC
```

```
DEBD3723 88FB664A
   8C7E6557 15DC5913 B44C7786 54AF32C0 AD5ADB28 98AF6B05
A3B197A5 EE73ED52
   38D8AF67 969D1E0E 505915A2 7228171E FA023118 14B017EB
02FA7E6D 9E16C8E3
   249C8524 F17BD8A9 82F70479 008CAEED 65F4E88B 182B3B72
713D0AD4 78B0EE88
   8F8A6180 EE0AF849 5E18EF1D E2CBCF7D 540B7D21 53C51938
3004EFB5 FB3134FD
   768B5790 1AAF5677 A53CEFAF 670AE0D4 20DC5742 4CDC6F1F
8A075139 D0CFBBC2
   89FD8528 F4F4D7CB 563E7CF7 7CD563F4
         quit

 license udi pid ISR4321/K9 sn FDO214328HZ
 no license smart enable
 diagnostic bootup level minimal

 spanning-tree extend system-id

 redundancy
 mode none

 interface Loopback0
 ip address 192.168.10.5 255.255.255.252
 ip ospf 2 area 1
 ipv6 address 201:DB8:ACAD:10::5/128
 ipv6 ospf 2 area 1

 interface GigabitEthernet0/0/0
 ip address 192.168.1.2 255.255.255.0
 ip ospf 2 area 1
 negotiation auto
 ipv6 address 2001:DB8:ACAD:1::2/64
 ipv6 ospf 2 area 1

 interface GigabitEthernet0/0/1
 ip address 192.168.2.1 255.255.255.0
 negotiation auto
 ipv6 address 2001:DB8:ACAD:2::1/64

 interface Serial0/1/0
 no ip address
 shutdown

 interface Serial0/1/1
 no ip address
```

```
shutdown

interface GigabitEthernet0/2/0
no ip address
shutdown
negotiation auto
interface GigabitEthernet0/2/1
no ip address
shutdown
negotiation auto

interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto

router ospf 2
router-id 2.2.2.2
redistribute connected subnets
redistribute bgp 1 subnets

router bgp 1
bgp log-neighbor-changes
neighbor 2001:DB8:ACAD:2::2 remote-as 2
neighbor 192.168.2.2 remote-as 2

address-family ipv4
  network 192.168.2.0
  redistribute connected
  redistribute ospf 2 metric 1
  no neighbor 2001:DB8:ACAD:2::2 activate
  neighbor 192.168.2.2 activate
exit-address-family

address-family ipv6
  redistribute connected
  redistribute ospf 2 metric 1
  network 2001:DB8:ACAD:2::/64
  neighbor 2001:DB8:ACAD:2::2 activate
exit-address-family

ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ip tftp source-interface GigabitEthernet0
```

```
 ipv6 router ospf 2
 control-plane

 line con 0
 transport input none
 stopbits 1
 line aux 0
 stopbits 1
 line vty 0 4
 login
```

Router 2

```
version 16.9
 service timestamps debug datetime msec
 service timestamps log datetime msec
 platform qfp utilization monitor load 80
 platform punt-keepalive disable-kernel-core
 hostname Router2
 boot-start-marker
 boot-end-marker
 vrf definition Mgmt-intf
 address-family ipv4
 exit-address-family
 address-family ipv6
 exit-address-family
 no aaa new-model
 login on-success log

 sbscriber templating
 vtp domain cisco
 vtp mode transparent
 ipv6 unicast-routing
 multilink bundle-name authenticated
 crypto pki trustpoint TP-self-signed-1411542050
 enrollment selfsigned
 subject-name cn=IOS-Self-Signed-Certificate-1411542050
 revocation-check none
 rsakeypair TP-self-signed-1411542050

 crypto pki certificate chain TP-self-signed-1411542050
 certificate self-signed 01
   30820330 30820218 A0030201 02020101 300D0609 2A864886
F70D0101 05050030
   31312F30 2D060355 04031326 494F532D 53656C66 2D536967
```

6E65642D 43657274
69666963 6174652D 31343131 35343230 3530301E 170D3235
31313132 31353238
35315A17 0D333030 31303130 30303030 305A3031 312F302D
06035504 03132649
4F532D53 656C662D 5369676E 65642D43 65727469 66696361
74652D31 34313135
34323035 30308201 22300D06 092A8648 86F70D01 01010500
0382010F 00308201
0A028201 0100E812 3ED4A890 B140F9BE 6AC4B04C 4F30D6C5
68AF7A43 A1A1EC22
880E97F9 94728900 F91F3878 C095AE84 51885992 F5EECD26
28918CCA A9B094C3
6EABD6D3 E25BAAA5 E6C06935 8C851CAC 46005A44 5A9AB1DE
D54281EE B3FE085A
9D3F3180 60E7851D 67314915 0C75C1E6 A38BD643 41C44159
3B503A3C 8C913CED
7378793C 7785548D 7A8411AB EB42ECB5 F7806873 68C2C19E
226C1DC4 92BAB47D
96562668 57DAB691 FAC9B558 1A1B9728 A3CAB237 B09BC76D
DE5BC448 9D2C6B03
A39B0EC2 DDBB1842 E7A06E44 E98CE99B 0606C709 9826F6D5
70FB767E 397E3F4C
9E412A16 B1106F1F 5E271AA8 BA27BC12 CCE3002C 25035F85
BA40F1B4 42588C36
2669DD77 C4610203 010001A3 53305130 0F060355 1D130101
FF040530 030101FF
301F0603 551D2304 18301680 148A9155 987F1601 1C7BBADA
BB3DF7A9 24233457
7A301D06 03551D0E 04160414 8A915598 7F16011C 7BBADABB
3DF7A924 2334577A
300D0609 2A864886 F70D0101 05050003 82010100 32ABE680
721B1755 41CFD94D
5748EB8C DA4B0535 1B3F9B9B 086F56D1 D4AC89F5 F0C7187A
27256D95 1AF7044B
F88C1465 D1C327BE BDE0B242 A9DA20A5 531E436C 732936BD
4DCC3A75 C43F27DA
32120909 63D0F978 158BC8BE 7F25F0E1 4D82DCD9 F169B62A
AF35BC37 97B52830
B99967E8 539B6994 D144E079 EFEB8143 387D0452 62905B55
B6F5D347 9649A3F2
3EE0E6F1 29B95A3A F8BC934B A20D8187 040253BB 1AD7D1AA
112DD801 46DB8355
5F367EE3 79A708FB B721A038 D0883A92 659CB25C 6D72884E
8886D337 51B1B1DC
1B62D52B 6AEC43AB 426631D3 602AAAD2 14617003 121C0F0A
718E09E6 C993C9B1

```
  774901C7 4BC04D37 F0DA5CC1 7B6D0174 8C2C3AC7
        quit

license udi pid ISR4321/K9 sn FDO215009PN
no license smart enable
diagnostic bootup level minimal

spanning-tree extend system-id

redundancy
mode none

interface Loopback0
ip address 10.0.0.9 255.255.255.255
ipv6 address 2001:DB8:ACAD:10::9/128
ipv6 eigrp 2

interface GigabitEthernet0/0/0
ip address 192.168.2.2 255.255.255.0
negotiation auto
ipv6 address 2001:DB8:ACAD:2::2/64

interface GigabitEthernet0/0/1
ip address 192.168.3.1 255.255.255.0
negotiation auto
ipv6 address 2001:DB8:ACAD:3::1/64
ipv6 eigrp 2

interface Serial0/1/0
no ip address
shutdown

interface Serial0/1/1
no ip address
shutdown

interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto

router eigrp 2
network 10.0.0.9 0.0.0.0
network 192.168.3.0
redistribute connected
redistribute bgp 2 metric 10000 100 255 1 1500
```

```
 eigrp router-id 3.3.3.3

 router bgp 2
 bgp log-neighbor-changes
 neighbor 2001:DB8:ACAD:2::1 remote-as 1
 neighbor 192.168.2.1 remote-as 1

 address-family ipv4
   network 192.168.2.0
   redistribute connected
   redistribute eigrp 2
   no neighbor 2001:DB8:ACAD:2::1 activate
   neighbor 192.168.2.1 activate
 exit-address-family

 address-family ipv6
   redistribute connected
   redistribute eigrp 2
   network 2001:DB8:ACAD:2::/64
   neighbor 2001:DB8:ACAD:2::1 activate
 exit-address-family

 ip forward-protocol nd
 ip http server
 ip http authentication local
 ip http secure-server
 ip tftp source-interface GigabitEthernet0

 ipv6 router eigrp 2
 redistribute bgp 2 metric 1000 100 255 1 1500
 redistribute connected

 control-plane

 line con 0
 transport input none
 stopbits 1
 line aux 0
 stopbits 1
 line vty 0 4
 login
```

Router 3

```
version 16.9
 service timestamps debug datetime msec
```

```
 service timestamps log datetime msec
 platform qfp utilization monitor load 80
 platform punt-keepalive disable-kernel-core
 hostname Router3
 boot-start-marker
 boot-end-marker
 vrf definition Mgmt-intf
 address-family ipv4
 exit-address-family
 address-family ipv6
 exit-address-family
 no aaa new-model
 login on-success log
 subscriber templating
 ipv6 unicast-routing
 multilink bundle-name authenticated
 crypto pki trustpoint TP-self-signed-3571946550
 enrollment selfsigned
 subject-name cn=IOS-Self-Signed-Certificate-3571946550
 revocation-check none
 rsakeypair TP-self-signed-3571946550
 crypto pki certificate chain TP-self-signed-3571946550
 certificate self-signed 01
   30820330 30820218 A0030201 02020101 300D0609 2A864886
F70D0101 05050030
   31312F30 2D060355 04031326 494F532D 53656C66 2D536967
6E65642D 43657274
   69666963 6174652D 33353731 39343635 3530301E 170D3235
31313132 31353533
   33325A17 0D333030 31303130 30303030 305A3031 312F302D
06035504 03132649
   4F532D53 656C662D 5369676E 65642D43 65727469 66696361
74652D33 35373139
   34363535 30308201 22300D06 092A8648 86F70D01 01010500
0382010F 00308201
   0A028201 0100DABE 7ABE7831 E8C07C1D 732C7489 514781C6
C1A8B63E FA798EF1
   0D48834E 8736D154 9DC222B9 5C107CA8 51B10A73 5BBD1F9E
2E25F8AB 4328FBFD
   816B96A5 8282EC26 4922C420 C6E7C68F 00C3C799 9C04F011
D34384FF 523343E9
   B004195D A3429431 0A59CF97 98CE5B09 2BA2C4F9 8802A46D
E7FFFE36 2922A1C7
   4A883832 70BD8A2B A3175DD9 2CCF576B 48615627 B651C6CA
C7658851 9E47C450
   3E8DA3A6 A34531EC A67A63EB 74080AE7 9BDD1BE7 4DD24B65
6C42A6DD A9DE646A
```

```
   A5B164E0 C799259E 8E8124B3 41117E66 759C8EEE FD202601
80D59A27 8FB0B331
   DB8E71D7 620EFCF2 F10D9519 E8690CC7 C57ACC85 12DA5341
D8C619A8 A3FEDE59
   146507DB 458D0203 010001A3 53305130 0F060355 1D130101
FF040530 030101FF
   301F0603 551D2304 18301680 14F09F26 40C06615 8B436A77
A4F85E5E E9F23A1D
   68301D06 03551D0E 04160414 F09F2640 C066158B 436A77A4
F85E5EE9 F23A1D68
   300D0609 2A864886 F70D0101 05050003 82010100 B9E1B6D7
5B12D3DD C26F52C9
   DDF02859 BB246DEB 240EE637 1B3EBAA9 79F6959E 941EA746
9453013F 884A7E1F
   8AEA2C65 91309948 DEF43A55 2B24A84F CF523F77 FFD4E1E2
7BB2664C 57E59A4D
   C76949F1 BCC928F4 1A43B0DE 47D757CF 21F729AF C758AFD6
811E1A67 1CB2BCFB
   C733C427 292DE37C F5BD53D4 6F21489D D7001B00 67D2BCCA
D460CC3F F2020FBB
   DAD53796 B2CD31EC A5BE2CD7 7C906B97 43FA727A C8074072
3A91B161 7FBC7FBD
   451F97CE 34C0E684 59B7A83B FD8B9BAA 05A87645 208214BD
968F7839 DE85912B
   3D3CEC62 F16F2611 57ACB732 2A0F6044 5676E3B1 DB192B0C
21505810 DBE30DB1
   D2BB2045 C84221DA 19CC53EB 77EEAFB8 520FC688
         quit
 license udi pid ISR4321/K9 sn FDO214420HS
 no license smart enable
 diagnostic bootup level minimal
 spanning-tree extend system-id
 redundancy
 mode none
 interface Loopback0
 ip address 10.0.0.13 255.255.255.252
 ipv6 address 2001:DB8:ACAD:10::13/128
 ipv6 eigrp 2
 interface GigabitEthernet0/0/0
 ip address 192.168.3.2 255.255.255.0
 negotiation auto
 ipv6 address 2001:DB8:ACAD:3::2/64
 ipv6 eigrp 2
 interface GigabitEthernet0/0/1
 ip address 192.168.4.1 255.255.255.0
 negotiation auto
 ipv6 address 2001:DB8:ACAD:4::1/64
```

```
interface Serial0/1/0
no ip address
interface Serial0/1/1
no ip address
interface GigabitEthernet0/2/0
no ip address
negotiation auto
interface GigabitEthernet0/2/1
no ip address
negotiation auto
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
negotiation auto
router eigrp 2
network 10.0.0.12 0.0.0.3
network 192.168.3.0
redistribute connected
redistribute bgp 2 metric 10000 100 255 1 1500
eigrp router-id 4.4.4.4
router bgp 2
bgp log-neighbor-changes
neighbor 2001:DB8:ACAD:4::2 remote-as 3
neighbor 192.168.4.2 remote-as 3
address-family ipv4
  network 192.168.4.0
  redistribute connected
  redistribute eigrp 2
  no neighbor 2001:DB8:ACAD:4::2 activate
  neighbor 192.168.4.2 activate
exit-address-family
address-family ipv6
  redistribute connected
  redistribute eigrp 2
  network 2001:DB8:ACAD:4::/64
  neighbor 2001:DB8:ACAD:4::2 activate
exit-address-family
ip forward-protocol nd
no ip http server
ip http secure-server
ipv6 router eigrp 2
redistribute connected
redistribute bgp 2 metric 10000 100 255 1 1500
control-plane

line con 0
transport input none
```

```
 stopbits 1
 line aux 0
 stopbits 1
 line vty 0 4
 login
```

Router 4

```
version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
platform punt-keepalive disable-kernel-cor
hostname router
boot-start-marker
boot-end-marker
vrf definition Mgmt-intf
address-family ipv4
exit-address-family
address-family ipv6
exit-address-family
no aaa new-model
login on-success log
subscriber templating
ipv6 unicast-routing
multilink bundle-name authenticated
crypto pki trustpoint TP-self-signed-187689846
enrollment selfsigned
subject-name cn=IOS-Self-Signed-Certificate-187689846
revocation-check none
rsakeypair TP-self-signed-187689846
crypto pki certificate chain TP-self-signed-187689846
certificate self-signed 01
  3082032E 30820216 A0030201 02020101 300D0609 2A864886 F70D0101
05050030
  30312E30 2C060355 04031325 494F532D 53656C66 2D536967 6E65642D
43657274
  69666963 6174652D 31383736 38393834 36301E17 0D323531 30323632
33353934
  325A170D 33303031 30313030 30303030 5A303031 2E302C06 03550403
1325494F
  532D5365 6C662D53 69676E65 642D4365 72746966 69636174 652D3138
37363839
  38343630 82012230 0D06092A 864886F7 0D010101 05000382 010F0030
```

```
82010A02
  82010100 B54A618D A8A9A76D A82A84ED A08F76AA 0E086D4C AD31CDC8
47A23BFA
  DE6C3C2E A4B0DE2F 7CAFD76E F9844D8D 34C13F0F 9F5CCBC5 8E25FA95
4A0EB758
  82D987D9 E6E21587 CA03EFC6 FF9B96E7 3EEFFFB7 26448FE9 C876C27C
3858D491
  511B51D5 7130AA34 EF696CA9 6E38C90F 3F39A945 72089B1E 0BFAE280
A5D498DB
  DEC854FE 7A0F2308 C42F0DCF DC716CB1 99A3F4CF 633BCB17 4076409F
7BBCA03C
  F0C92991 53401DD1 463526E9 06F343E3 46459FC1 AC519244 51E3144F
53E59353
  CE6006B9 D5E9B4CA 797100A8 E7029E4B 2A95B0CC 34697676 25482E70
B17F75C2
  3B75B865 53066012 E75D39C0 F9385A0A 86DA63C7 6577B5F0 D31B6C6C
CC0153C8
  7EB766F9 02030100 01A35330 51300F06 03551D13 0101FF04 05300301
01FF301F
  0603551D 23041830 168014E0 218801BC 65C4EB28 13A9C418 9D10E65A
3487A430
  1D060355 1D0E0416 0414E021 8801BC65 C4EB2813 A9C4189D 10E65A34
87A4300D
  06092A86 4886F70D 01010505 00038201 01003EDC 95C3B485 845F7530
0B7EAF5F
  B588AB9D BB5ED008 CB094146 E712C3B4 5B07B163 F9C25330 3CFB1558
70F67F38
  92675D3F 6139744D 78655870 D301E2CA 04AA6335 3ADAB00A 52042B57
40AEF043
  47FB44F9 A91AE3EE F859D535 A5E8E7B4 31E75092 A0634EFA DBA58F2D
FB171D3A
  49D5BFDE 630E6127 EECDDE68 32E4E39A 5208D1F9 4AD67821 C91F825C
8D25247D
  64B764A6 054BCE86 8F1A1430 3B354515 E2721E61 DC2372EB 1D0DA364
B36C6D1D
  64BB0012 5D5212D8 6E15262A EFBC9F9D 53EA9D86 F298F48C 620439CB
B477E4E0
  3578C4D4 F648B431 861B4A2D 83A967EE 13CC380B F9A2C1D7 1169894C
B7997EE2
  ACF65F5C F902A5B0 78FEB830 BD26C524 9EC3
        quit

license udi pid ISR4321/K9 sn FLM2407014H
no license smart enable
diagnostic bootup level minimal
spanning-tree extend system-id
redundancy
```

```
mode none
interface Loopback0
ip address 10.0.0.17 255.255.255.252
ip router isis 2
ipv6 address 2001:DB8:ACAD:10::17/128
ipv6 router isis 2
interface GigabitEthernet0/0/0
ip address 192.168.4.2 255.255.255.0
negotiation auto
ipv6 address 2001:DB8:ACAD:4::2/64
interface GigabitEthernet0/0/1
ip address 192.168.5.1 255.255.255.0
ip router isis 2
negotiation auto
ipv6 address 2001:DB8:ACAD:5::1/64
ipv6 router isis 2
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto
router isis 2
net 02.0001.0000.0000.0001.00
metric-style narrow
redistribute connected
redistribute bgp 3
router bgp 3
bgp log-neighbor-changes
neighbor 2001:DB8:ACAD:4::1 remote-as 2
neighbor 192.168.4.1 remote-as 2
address-family ipv4
  network 192.168.4.0
  redistribute connected
  redistribute isis 2 level-1-2 metric 2
  no neighbor 2001:DB8:ACAD:4::1 activate
  neighbor 192.168.4.1 activate
exit-address-family
address-family ipv6
  redistribute connected
  redistribute isis 2 metric 2 level-1
  network 2001:DB8:ACAD:4::/64
  neighbor 2001:DB8:ACAD:4::1 activate
exit-address-family
ip forward-protocol nd
ip http server
ip http authentication local
```

```
ip http secure-server
ip tftp source-interface GigabitEthernet0
 control-plane
line con 0
transport input none
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
end
```

Router 5

```
version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
platform punt-keepalive disable-kernel-core
hostname Router5
boot-start-marker
boot-end-marker
vrf definition Mgmt-intf
address-family ipv4
exit-address-family
address-family ipv6
exit-address-family
no aaa new-model
login on-success log
subscriber templating
ipv6 unicast-routing
multilink bundle-name authenticated
crypto pki trustpoint TP-self-signed-1541607431
enrollment selfsigned
subject-name cn=IOS-Self-Signed-Certificate-1541607431
revocation-check none
rsakeypair TP-self-signed-1541607431
crypto pki certificate chain TP-self-signed-1541607431
certificate self-signed 01
  30820330 30820218 A0030201 02020101 300D0609 2A864886 F70D0101
05050030
  31312F30 2D060355 04031326 494F532D 53656C66 2D536967 6E65642D
43657274
  69666963 6174652D 31353431 36303734 3331301E 170D3235 31313132
31363137
  34305A17 0D333030 31303130 30303030 305A3031 312F302D 06035504
```

```
03132649
  4F532D53 656C662D 5369676E 65642D43 65727469 66696361 74652D31
35343136
  30373433 31308201 22300D06 092A8648 86F70D01 01010500 0382010F
00308201
  0A028201 0100B06D FAB2B6CC BD1D0E97 1DF4438F 26BD847A DF5CB294
3720F8C6
  19AAD5DF 705CC206 350113AC 99D8069C BB5FC84C 36DD345C 46AF7B35
97C48001
  267599E2 3BB50D80 7E8FB4D3 C81161D4 5E0BEF46 D941FAD3 27A8204B
3C7FE3A8
  B251269B 6E1B8A24 0ADB56B7 A2926F2A 167579ED EE1626C3 A16A98D9
DF152D49
  813888E0 9DB30C02 565BFD5A 1FC74283 04E81C65 E11D98E1 32A91EA4
29E02E64
  CD79F1FE 0BC89700 EBACB640 85901086 2514DEA1 8F6A09E2 932E8299
F2916EC7
  C72F211D E5A3B560 14DDB533 97C292EE 9FF00D6C BBDB996F FC8B239D
9065E823
  75C40A7D 27B64A99 711EE035 7FE44FCA A0993E37 58B6C1B8 937EE2AF
5A5ED000
  B1F5D026 4F2B0203 010001A3 53305130 0F060355 1D130101 FF040530
030101FF
  301F0603 551D2304 18301680 14AF282B 96251826 B7DAF29F D604FD16
2488E853
  11301D06 03551D0E 04160414 AF282B96 251826B7 DAF29FD6 04FD1624
88E85311
  300D0609 2A864886 F70D0101 05050003 82010100 4A48B9E6 F46FC5DE
7AE7E5C8
  02E5C0BA 00D32E30 2F4B6100 27358D3D BE54C7B6 6814D702 CE095D3B
45B29508
  6FFD2791 16A05B6C 96AF8835 F706E94A 23F50A90 447D10CB 2BDB810B
69A196EA
  263E4956 A62B776D 73E63ACA 76C66E18 373F7F02 7030A966 EFD45506
5020E155
  230C4CCD F73BB396 A5E94B2E 5BDE5D52 2B409B4B F4891C69 34ADC4B5
14FDCE55
  DB8B05EC 5241ADAC 38ACF6BB 36F49CD2 E3314E61 9FC1C045 D7934B16
4449A96A
  9C458445 5043F39F 3D78B7E4 E6D3F9FC 17D1A15F 64F72E83 C7051001
11F0D7A0
  0B29F222 F4B13704 7649E4DA A7303405 0119C8BF 4BDE7A10 E5F4E7B5
44FA393B
  383181E6 F94C981E 828C39C4 72248F36 962B3077
        quit
license udi pid ISR4321/K9 sn FDO214420GM
no license smart enable
```

```
diagnostic bootup level minimal
spanning-tree extend system-id
redundancy
mode none
interface Loopback0
ip address 10.0.0.21 255.255.255.252
ip router isis 2
ipv6 address 2001:DB8:ACAD:10::21/128
ipv6 router isis 2
interface GigabitEthernet0/0/0
ip address 192.168.5.2 255.255.255.0
ip router isis 2
negotiation auto
ipv6 address 2001:DB8:ACAD:5::2/64
ipv6 router isis 2
interface GigabitEthernet0/0/1
no ip address
shutdown
negotiation auto
interface Serial0/1/0
no ip address
shutdown
interface Serial0/1/1
no ip address
shutdown
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto
router isis 2
net 02.0001.0000.0000.0002.00
metric-style narrow
router isis
metric-style narrow
ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ip tftp source-interface GigabitEthernet0
control-plane
line con 0
transport input none
stopbits 1
line aux 0
stopbits 1
```

```
line vty 0 4
login
```

show commands

```
MLS1#traceroute 192.168.5.2
Tracing the route to 192.168.3.6
VRF info: (vrf in name/id, vrf out name/id)
  1 192.168.1.2 1 msec 1 msec 0 msec
  2 192.168.2.1 1 msec 2 msec 1 msec
  3 192.168.2.2 1 msec 1 msec 1 msec
  4 192.168.3.1 1 msec 2 msec 1 msec
  5 192.168.3.2 1 msec 2 msec 1 msec
  6 192.168.4.1 2 msec 1 msec 1 msec
  7 192.168.4.2 0 msec 1 msec 2 msec
  8 192.168.5.1 2 msec 2 msec 1 msec
  9 192.168.5.2 0 msec 1 msec *

R5#traceroute 192.168.1.1
Tracing the route to 192.168.1.1
VRF info: (vrf in name/id, vrf out name/id)
  1 192.168.5.1 2 msec 1 msec 0 msec
  2 192.168.4.2 0 msec 1 msec 1 msec
  3 192.168.4.1 1 msec 1 msec 1 msec
  4 192.168.3.2 2 msec 1 msec 1 msec
  5 192.168.3.1 2 msec 2 msec 1 msec
  6 192.168.2.2 2 msec 1 msec 1 msec
   7 192.168.2.1 0 msec 2 msec 1 msec
  8 192.168.1.2 2 msec 0 msec 1 msec
  9 192.168.1.1 0 msec 1 msec *
```

R1:

```
     10.0.0.13/28 is subnetted, 1 subnets
O E2     3.3.3.3 [110/1] via 192.168.1.2, 23:10:38, GigabitEthernet0/0/0
     10.0.0.17/28 is subnetted, 1 subnets
O E2     4.4.4.4 [110/1] via 192.168.1.2, 23:10:09, GigabitEthernet0/0/0
     10.0.0.21/28 is subnetted, 1 subnets
O E2     5.5.5.5 [110/1] via 192.168.1.2, 23:04:38, GigabitEthernet0/0/0
     10.0.0.25/28 is subnetted, 1 subnets
O E2     6.6.6.6 [110/1] via 192.168.1.2, 00:07:47, GigabitEthernet0/0/0
```

```
      192.168.1.0/24 is variably subnetted, 3 subnets, 2 masks
C        192.168.1.0/30 is directly connected, GigabitEthernet0/0/0
L        192.168.1.1/32 is directly connected, GigabitEthernet0/0/0
O E2     192.168.1.4/30
           [110/20] via 192.168.1.2, 23:10:50, GigabitEthernet0/0/0
      192.168.2.0/30 is subnetted, 1 subnets
O E2     192.168.2.0 [110/1] via 192.168.1.2, 23:10:38,
GigabitEthernet0/0/0
      192.168.3.0/30 is subnetted, 1 subnets
O E2     192.168.4.1 [110/1] via 192.168.1.2, 00:08:17,
GigabitEthernet0/0/0
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Problems

This lab was a very difficult lab because of the research and knowledge that was required to configure BGP. There were many different things that would not work because BGP was misconfigured or we incorrectly used a command. We would have to continuously go back and reconfigure routers because of incorrect configurations. We commonly could not redistribute BGP routes through the network. We were able to get ipv4 up after further research and understanding of BGP. After getting more experience with BGP we familiarized ourselves with the BGP commands and reconfigured. Ipv6 also became a challenge to us because the commands for many routing protocols as well as BGP are different. We used address families because they help to separate the configurations and are required for ipv6 connectivity. If I was to do this lab again with the knowledge that I had I would spend time researching BGP and example topologies and configurations in order to familiarize myself with BGP and the required configs.

We also had common issues with ISIS because it is relatively new to us. We should have familiarized ourselves with ISIS before starting the lab. This would have caused the lab to go smoother if we knew how ISIS worked better.

After finishing our ipv4 configurations we left and came back the next day the power had gone out which had wiped all of our configs. This basically set us all the way back and wasted a lot of time. While we were able to reconfigure our BGP quicker the second time it also allowed to understand what was configured on all the routers better and guaranteed that the commands were at least used correctly. Despite thinking that this would solve the issues our IPv6 still would not ping through the network. A few commands that were meant for ipv4 BGP were used for ipv6 and would obviously not work. After some more looking we were able to finally get both ipv4 and ipv6 to ping through the network.

We also configured two extra attributes with BGP: weight and local preference. When I was configuring both of them, I configured them on the same router. When both did not work on the same router we immediately checked to see if the route map as the issue because we figured that the route map was the only thing that could have an issue in the configuration. After checking multiple things and not seeing any issues in the configuration, we decided to do a bit more research and realized that both can not be configured on the same router. When we changed this in immediately worked.

Overall, the configurations for this lab were very complicated and allowed for many issues that were very difficult to troubleshoot as there could have been may different things that would cause connectivity issues. BGP, while very useful, is also very easy to incorrectly configure.

Conclusion

When doing BGP we learned that when doing a much larger lab and configuration like this one that planning, making a brief outline, and researching are crucial for time management and efficiency. Documenting and saving configurations are things that we hadn’t done as much of in the past which slowed us down in this lab. We also learned the basics of BGP and where it should be used in a network and what it can do if incorrectly used.

layer 2 attacks lab

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